1.1. The project summary

Project Number [1]	958402	Project Acronym [2]	AI-CUBE

One form per project	
General information	
Project title [3]	Artificial Intelligence and Big Data CSA for Process Industry Users, Business Development and Exploitation
Starting date [4]	01/09/2020
Duration in months [5]	24
Call (part) identifier [6]	H2020-NMBP-ST-IND-2020-singlestage
Topic	DT-SPIRE-11-2020 Artificial Intelligence and Big Data Technologies for Process Industries (CSA)
Fixed EC Keywords	Artificial intelligence, intelligent systems, multi agent systems, Smart manufacturing
Free keywords	

Abstract [7]

AI-CUBE seeks to enhance the understanding of different digital technologies related to artificial intelligence (AI) and big data (BD) applied in process industries for all the SPIRE industrial sectors (cement, ceramics, chemicals, engineering, minerals and ores, non-ferrous metals, steel, water). Therefore, a close collaboration with industry is mandatory to achieve in-depth insights into possible application areas of AI for processes, technology, sensor applicability and assessment of their level of penetration.

The overall project approach is based on the development of a 3-dimensional conceptual matrix based on:

1) AI and BG technologies
2) Application areas (activities and industrial processes)
3) SPIRE sectors

AI-CUBE's main goal is to define a roadmap in AI and the use of BD for the process industry and their maturity level across the industrial sectors, including guidelines for implementation. Industrial stakeholders and associations will validate the consolidated roadmap ensuring solution feasibility and benefits for the European industrial community. A crosslinked vision over process industry sectors shall facilitate cooperation and boost technologies deployment at their full potential. An in-depth consultation with industry (association, representatives, companies) will provide an overview of current AI and BD algorithms application, identifying exploitable synergies among sectors. A deep study of the

application areas in planning and operations within other industrial sectors facilitates a gap analysis, propitiating knowledge sharing among processes and sectors.

A Multi-Actor Multi-Criteria analysis will obtain a widely supported and consensus-based action plan for industrial consultation. This will allow the inclusion of a broad stakeholder community representing the main industry actors throughout all the SPIRE sectors, with which the project consortium has strong connections that will support sector integration and stakeholders' engagement.

1.2. List of Beneficiaries

Project Number [1]	958402	Project Acronym [2]	AI-CUBE

List of Beneficiaries

No	Name	Short name	Country	Project entry month[8]	Project exit month
1	CIAOTECH Srl	PNO	Italy	1	24
2	FUNDACION ZARAGOZA LOGISTICS CENTER	ZLC	Spain	1	24
3	FRAUNHOFER GESELLSCHAFT ZUR FOERDERUNG DER ANGEWANDTEN FORSCHUNG E.V.	Fraunhofer	Germany	1	24
4	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	IRIS	Spain	1	24
5	CONSIGLIO NAZIONALE DELLE RICERCHE	CNR	Italy	1	24

1.3. Workplan Tables - Detailed implementation

1.3.1. WT1 List of work packages

WP Number[9]	**WP Title**	**Lead beneficiary**[10]	**Start month**[12]	**End month**[13]
WP1	Current landscape analysis: AI & BD technologies and industry applications	4 - IRIS	1	7
WP2	Industry consultation and AI&BD framework for maturity level	5 - CNR	2	24
WP3	Mapping of AI & BD technologies and maturity level assessment	3 - Fraunhofer	8	19
WP4	Roadmap and industry synergies	2 - ZLC	12	24
WP5	Dissemination, Communication and Exploitation	1 - PNO	1	24
WP6	Project coordination and quality management	1 - PNO	1	24

TASK/WP	TASK/WP TITLE	TASK LEADER	START MONTH	END MONTH	DURATION IN MONTHS	YEAR 1												YEAR 2											
						1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
WP1	**Current landscape analysis: AI & BD technologies and industry applications**	**IRIS**	**M1**	**M7**	**7**																								
T1.1	Identification of AI & BD technologies	IRIS	M1	M4	4																								
T1.2	Industrial processes analysis per sector and company size	IRIS	M3	M6	4																								
T1.3	Macro applications areas plan review and update	IRIS	M5	M7	3																								
WP2	**Industry consultation and AI&BD framework for maturity level**	**CNR**	**M5**	**M12**	**8**																								
T2.1	Stakeholders interviews and workshops	CNR	M5	M9	5																								
T2.2	Framework for the AI&BD technologies maturity level assessment	CNR	M6	M9	4																								
T2.3	Assessment of impact of AI&BD	CNR	M7	M12	6																								
WP3	**Mapping of AI & BD technologies and maturity level assessment**	**IML**	**M8**	**M18**	**11**																								
T3.1	Compilation of AI & BD technology per application and sector	IML	M8	M12	5																								
T3.2	Mapping AI & BD technologies per sector and macro area	IML	M11	M18	8																								
T3.3	Methods for the exploitation of AI methods	IML	M15	M17	3																								
T3.4	Industry validation	IML	M16	M19	3																								
WP4	**Roadmap and industry synergies**	**ZLC**	**M12**	**M24**	**10**																								
T4.1	Gaps and opportunities analysis – recommendations	ZLC	M15	M18	4																								
T4.2	CASE STUDIES – Business models	ZLC	M12	M23	8																								
T4.3	Development of the roadmap per sector and area	ZLC	M17	M24	8																								
T4.4	New paths of collaboration for ELSI	ZLC	M20	M24	5																								
T4.5	Validation	ZLC	M20	M24	5																								
WP5	**Dissemination, Communication and Exploitation**	**PNO**	**M1**	**M24**	**24**																								
T5.1	Definition of the strategy for dissemination of project results	PNO	M1	M3	3																								
T5.2	Communication activities	PNO	M1	M24	24																								
T5.3	Exploitation and dissemination	PNO	M1	M24	24																								
WP6	**Project coordination and quality management**	**PNO**	**M1**	**M24**	**24**																								
T6.1	Consortium and project administrative management	PNO	M1	M24	24																								
T6.2	Monitor technical Developments and Work Plan	PNO	M1	M24	24																								
T6.3	Quality Management	PNO	M1	M24	24																								

1.3.2. WT2 list of deliverables

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.1	AI & BD technologies state-of-the-art review	WP1	4 - IRIS	Report	Public	4
D1.2	Sectorial analysis of industrial processes in Europe	WP1	4 - IRIS	Report	Public	6
D1.3	Review and update of the identified macro applications areas plan	WP1	4 - IRIS	Report	Public	7
D2.1	Plan for stakeholder involvement	WP2	1 - PNO	Report	Confidential, only for members of the consortium (including the Commission Services)	2
D2.2	Report on implemented collaboration mechanisms	WP2	1 - PNO	Report	Public	24
D2.3	Framework for AI & BD maturity level assessment	WP2	5 - CNR	Report	Public	8
D2.4	Report on the assessment of impact	WP2	5 - CNR	Report	Public	12
D3.1	Outline on applied AI methods & BD technology per sector	WP3	3 - Fraunhofer	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D3.2	Cross-comparison between industries and possible areas of transfer	WP3	3 - Fraunhofer	Report	Confidential, only for members of the consortium (including the Commission Services)	17
D3.3	Overview on validated methods per sector and validated methods for cross-sectoral and cross-industrial transfer	WP3	1 - PNO	Report	Confidential, only for members of the consortium (including the Commission Services)	19

D4.1	Business Model Game development	WP4	1 - PNO	Demonstrator	Public	18
D4.2	Gaps and opportunities analysis – recommendations and business models	WP4	2 - ZLC	Report	Public	18
D4.3	AI-CUBE roadmap	WP4	2 - ZLC	Report	Public	24
D4.4	New paths of collaboration for ELSI	WP4	5 - CNR	Report	Public	24
D5.1	Dissemination Plan	WP5	1 - PNO	Report	Public	3
D5.2	Website and AI-CUBE dissemination and exploitation platform	WP5	1 - PNO	Websites, patents filling, etc.	Public	4
D5.3	A - Report on Dissemination, Communication & Exploitation activities	WP5	1 - PNO	Report	Public	12
D5.4	B - Report on Dissemination, Communication & Exploitation activities	WP5	1 - PNO	Report	Public	24
D6.1	Project Management Plan	WP6	1 - PNO	Report	Public	2
D6.2	Quality Management Plan	WP6	1 - PNO	Report	Public	2
D6.3	Data management Plan – DMP	WP6	2 - ZLC	ORDP: Open Research Data Pilot	Public	6

1.3.3. WT3 Work package descriptions

Work package number [9]	WP1	**Lead beneficiary** [10]	4 - IRIS
Work package title	Current landscape analysis: AI & BD technologies and industry applications		
Start month	1	**End month**	7

Objectives

The main objective of WP1 is to establish the current technological and industrial landscape regarding AI and BD technologies and process industry sectors in Europe, setting the basis of the following WPs work in defining mapping tools and the roadmap for AI and BD. In order to properly do so, literature reviews and consultation with relevant stakeholders will start from the very beginning of the project. Furthermore, a detailed assessment of both the technological and industrial status will allow to adjust the original implementation plan (if needed), expanding it to additional macro applications areas.

Description of work and role of partners

WP1 - Current landscape analysis: AI & BD technologies and industry applications [Months: 1-7]
IRIS, PNO, ZLC, Fraunhofer, CNR
T1.1 Mapping of AI & BD technologies (M 1-4)
(Lead IRIS, contribution CNR, PNO)
An overview of the AI and BD existing technologies will be generated by means of literature review, consultation with ICT community, analysis of existing roadmaps at different levels (regional, national, and international), as well as focused intelligence activity for the identification of the most important research and innovation projects of H2020 and other EU framework Programmes. A comparison between roadmaps and running projects will give a preliminary picture of how technologies are under implementation and how projects already cover objectives set by roadmaps and strategic research agendas.
A first overall description of the technologies key features, maturity level and applicability will be outlined, providing an adequate starting point for the detailed mapping outlined in WP3 and the roadmap for process industry established in WP4.
T1.2 Industrial processes analysis per sector and company size (M 3-6)
(Lead IRIS, contribution PNO)
A general review regarding the most relevant processes of the key industry sectors represented by SPIRE (i.e. cement, ceramics, chemicals, non-ferrous metals, minerals, steel and water) will be developed. Process and product-oriented stages such as R&D, design and production will be addressed, but also marketing, sourcing and distribution will be taken into consideration. The review will also be categorised by dividing the needs depending on company size/resources. Once a first shortlist is generated, it will be shared within the SPIRE community for complementary feedback, aiming at guaranteeing proper representativeness of process industry reality throughout Europe.
Specific challenges related to each of these processes will also be taken into consideration in order to drive the subsequent analysis. Coming from the five “macro” areas predefined during the preparation stage, a shortlist of industrial tasks or activities which could benefit for the deployment of these technologies will be prepared. Special focus will be given to existing AI and BD applications in process industries nowadays and the leaders for each sector and company size, but other transversal applications coming from other sectors will also be reviewed. The main aim of this action will be to define the key bottlenecks and real problems for AI and BD strategies uptake.
As a complement to this first industrial analysis, a list of key industrial stakeholders will be identified through a focused analysis of the stakeholders showing direct connection to the consortium companies and other relevant actors in the EU scenario. Moreover, possible stakeholders meeting events will be drafted to set the basis of the industrial consultation process to be developed in WP2.
T1.3 Macro applications areas plan review and update (M 5-7)
(Lead IRIS, contribution CNR. ZLC, IML)
By taking into consideration the information gathered in tasks T1.1 and T1.2, a plan update for the subsequent WP of the project will be outlined, focusing on the AI and BD technologies application in the five “macro” areas. An extended and complementary literature review will be done to assure that the following areas are adequately assessed, while any other area not identified until this moment could also be taken into consideration:
• Research and innovation management, planning, and design (e.g., new chemical synthesis strategies, health and safety assessments);
• Process control: yield and accuracy enhancement;

- Supply chain management and scheduling of connected processes, plants and/or sites (e.g., for industrial symbiosis),process flexibility;
- Predictive maintenance;
- Product customisation and product traceability.

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.1	AI & BD technologies state-of-the-art review	4 - IRIS	Report	Public	4
D1.2	Sectorial analysis of industrial processes in Europe	4 - IRIS	Report	Public	6
D1.3	Review and update of the identified macro applications areas plan	4 - IRIS	Report	Public	7

Description of deliverables

D1.1 AI & BD technologies state-of-the-art review (M4) (IRIS). Report describing the current status of AI and BD technologies and their application degree in industrial environments.
D1.2 Sectorial analysis of industrial processes in Europe (M6) (IRIS). Report summarising the most relevant industrial processes per process industry sector in Europe, highlighting major challenges where AI and BD could play a relevant role.
D1.3 Review and update of the identified macro applications areas plan (M7) (IRIS). Review and update of the proposed plan for the consolidated macro applications areas in process industry.

D1.1 : AI & BD technologies state-of-the-art review [4]

Report describing the current status of AI and BD technologies and their application degree in industrial environments.

D1.2 : Sectorial analysis of industrial processes in Europe [6]

Report summarising the most relevant industrial processes per process industry sector in Europe, highlighting major challenges where AI and BD could play a relevant role

D1.3 : Review and update of the identified macro applications areas plan [7]

Review and update of the proposed plan for the consolidated macro applications areas in process industry.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS1	Map and Analysis of AI&BD technologies in industrial processes	4 - IRIS	18	AI&BD technologies mapped and analysed per industrial processes in SPIRE sectors

Work package number [9]	WP2	**Lead beneficiary** [10]	5 - CNR
Work package title	Industry consultation and AI&BD framework for maturity level		
Start month	2	**End month**	24

Objectives

The goal of the work package is twofold: i) set a strategy to work in a continuous way with the stakeholders and ii) design a framework to categorize current AI and BD technologies' applications in consideration of the actual use and knowledge of their application possibilities in industrial environments. This step should enable the identification of the business expectation of the different companies and prepare the work of WP3 related to the assessment and applicability.
Moreover, this framework will take into consideration also the ELSI dimension as a layer of the model itself.

Description of work and role of partners

WP2 - Industry consultation and AI&BD framework for maturity level [Months: 2-24]
CNR, PNO, ZLC, Fraunhofer, IRIS
T2.1 Strategies for stakeholders' engagement (M 5-9)
(Lead PNO, contribution CNR)
The stakeholder engagement will be driven and structured by the Multi-Actor Multi-Criteria Analysis (MAMCA) methodology38, which will be used to consult a broad stakeholder community representing the main sectors of the SPIRE industry in Europe. This stepwise and scientifically sound approach allows the consortium of the project to involve a large group of stakeholders in the process of analysis if the AI and BD technologies implemented in the SPIRE industry.
On the basis of each partner network of contacts and the MAMCA methodology – as a starting point - the task aims to define a collaboration mechanism to reinforce the fully functional, well connected and lively community of SPIRE at European level establishing also a close collaboration with the ICT community.
A strategy for networking creation and management will be jointly developed and implemented. Involvement of stakeholders will be defined according to the specific stage of development of the project and according to the planned tasks. Collaboration mechanisms will be defined in accordance with WP5 and some specific dissemination means will be defined in order to have the maximum benefit for the project and the stakeholders in a win-win approach.
During the project duration, workshops (in the form of focus group, brainstorming session etc.), and interviews will be organized aiming to gather input from the large community of SPIRE. Also, major AI and BD providers will be engaged and involved along the project to help the cross-cutting activities to merge visions from the point of view of demand and offer of AI and BD. Workshops will be organised to bring together each time a number of stakeholders (25-50 participants per workshop) that can be easily managed with 1 or 2 working sessions to collect the required input from the participants related to the specific stage of the methodology. Direct interviews with SPIRE members and managers from industries and AI and BD providers will be organized with the aim to identify pilot cases and emphasize their innovativeness in the process industry landscape: all the partners will be involved and in charge to identify in their own network interesting industrial cases.
Each partner will be also in charge to organise at least one of such workshops in order to assure as much as possible coverage of European countries as well as to involve different actors of the value chain. Besides gathering input for the different activities of AI-CUBE, these workshops will also serve as networking among stakeholders with a crosssectorial approach to take the opportunity to fertilize practices from one sector to another. Some possible organizations which are already in the network of the partners and can be easily involved for the workshops are:
Cement: Cemex Research Group AG
Ceramics: Keraben Grupo S.A.; AL-FARBEN, S.A.
Chemicals: CEFIC, DOW Chemical. P&G, BAYER, BASF, Evonik, Covestro, Clariant; Repsol, Foresa, BASF Non-Ferrous metals: Stena
Minerals: LCM, Part of Great Western Minerals Group,
Steel: Tenova, Tenaris, VDEh-Betriebsforschungsinstitut GmbH; K1-MET GmbH; DK Recycling Und Roheisen Gmbh Engineering: YARA
Water: ECOLAB – NALCO, European Water Platform, DOW chemical, Veolia, ZINNAE
AI and BD actors: BDVA (Big Data Value Association), AIXIA (Italian Association for Artificial Intelligence), SAP, Siemens

A preliminary list of meetings with stakeholders is here proposed. Some of these workshops shall be organized in conjunction with other events at European level in order to assure larger audience and to optimize the costs.
Per each meeting, the following list indicates the respective: Task; type if Meeting with experts and stakeholders; date; responsible partner; place.
- T1.2; Workshop on industrial processes - M3-4 (IRIS) Brussels
- T2.2-T2.3 ; Restricted workshop with selected experts for validation of the framework - M8 (STIIMA) Italy
- T3.1: Interviews, workshops and focus groups - M9-12 (IML) Various
- T3.4: Workshop for validation - M18 (IML) Dortmund
- T4.1: Workshop on gap analysis - M16 (ZLC) Zaragoza
- T4.2: Interviews and focus groups - M18-20 (ZLC) Various
- T4.5: Workshop for validation of the roadmap - M23 (ZLC) Brussels

T2.2 Framework for AI and BD maturity level assessment (M 6-9)
(Lead CNR, contribution all)
The objective of this task is to define a conceptual framework for the evaluation of the maturity level of process industry in the application of AI and BD. Once in WP1 the categorization of the different types of AI and BD are defined, it is necessary to identify different levels of application of each AI and BD technology taking into consideration processes and activities. In literature there are already available some systems to evaluate the overall digital maturity level where it is possible to obtain the status of a company in terms of digitalisation with a benchmark comparison. These systems will be analysed and a new framework specifically for AI and BD will be developed to: (i) provide an impartial and complete evaluation system updated with the state-of-the-art; (ii) understand the industrial needs in terms of technologies (iii) identify a cross-sectorial comparative system to define maturity level and furthermore suggest new actions for improvement. The framework will take into consideration not only technological maturity (TRL) and a special session of the model will consider also criteria to evaluate maturity towards ELSI (Ethical, Legal, Social Issues). This framework will be used and applied in WP3.

T2.3 Assessment of impact of AI and BD (M 7-12)
(Lead ZLC, contribution CNR, IML)
This task is dedicated to the assessment of the impact of AI and BD on the business processes expected by companies according to some dimensions like: • Impact on human/operators
• Impact on Process/Product RD&I
• Impact on Plant and Plant Operations
• Impact on Value Chains
• Impact on organisation in factories and in companies
The task will work on the basis of the preliminary impact indicators identified in the SPIRE document on AI and will further extrapolate a more detailed analysis of the expectation of the companies in terms of performance (cost, agility, responsiveness, etc).
This task will be supported by the use of several different methodologies among which Multi-Actor Multi-Criteria analysis, AHP and others as a way to weight in a comparative way the contribution and the input of several stakeholders.

List of deliverables

Deliverable Number[14]	**Deliverable Title**	**Lead beneficiary**	**Type**[15]	**Dissemination level**[16]	**Due Date (in months)**[17]
D2.1	Plan for stakeholder involvement	1 - PNO	Report	Confidential, only for members of the consortium (including the Commission Services)	2
D2.2	Report on implemented collaboration mechanisms	1 - PNO	Report	Public	24

D2.3	Framework for AI & BD maturity level assessment	5 - CNR	Report	Public	8
D2.4	Report on the assessment of impact	5 - CNR	Report	Public	12

Description of deliverables

D2.1: Plan for stakeholder involvement (M2) (PNO). Plan with strategies to involve stakeholders along the project activities in T2.1
D2.2: Report on implemented collaboration mechanisms (M24) (PNO). Detailed summary of all initiatives put in place with stakeholders, minutes of the meetings, interviews etc. as from T2.1.
D2.3: Framework for AI&BD maturity level assessment (M8) (CNR). Description of the framework for the maturity assessment developed in T2.2.
D2.4: Report on the assessment of impact (M12) (CNR). Report on the development of a system for assessing impact of AI&BD as developed in T2.3.

D2.1 : Plan for stakeholder involvement [2]

Plan with strategies to involve stakeholders along the project activities in T2.1

D2.2 : Report on implemented collaboration mechanisms [24]

Detailed summary of all initiatives put in place with stakeholders, minutes of the meetings, interviews etc. as from T2.1.

D2.3 : Framework for AI & BD maturity level assessment [8]

Description of the framework for the maturity assessment developed in T2.2.

D2.4 : Report on the assessment of impact [12]

Report on the development of a system for assessing impact of AI&BD as developed in T2.3.

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS1	Map and Analysis of AI&BD technologies in industrial processes	4 - IRIS	18	AI&BD technologies mapped and analysed per industrial processes in SPIRE sectors
MS2	Consultation with stakeholders	1 - PNO	9	Stakeholders engagement
MS3	Assessment framework defined	5 - CNR	9	Definition of the criteria to describe and assess the

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
				AI&BD technology level of penetration and maturity.

Work package number [9]	WP3	**Lead beneficiary** [10]	3 - Fraunhofer
Work package title	Mapping of AI & BD technologies and maturity level assessment		
Start month	8	**End month**	19

Objectives

The main goal of WP3 is the categorization of the current AI and BD technologies taking into consideration the actual level of maturity regarding their application possibilities in industrial environments. The list of the individual technologies would enable the systematic identification and further development of application areas and possible uses of AI and BD technologies in the process industry, in accordance to their current level of penetration.

Description of work and role of partners

WP3 - Mapping of AI & BD technologies and maturity level assessment [Months: 8-19]
Fraunhofer, PNO, ZLC, IRIS, CNR
T3.1 Compilation of AI & BD technology per application and sector (M 8-12)
(Lead IML, contribution CNR)
Within this work package current AI & BD technologies are identified and clustered according to their application areas and fields of use. The term AI includes applications such as modelling of static and dynamic systems, decision support systems or applications in predictive maintenance. The use of Big Data (BD) is in many areas a basic requirement for the handling of AI technologies, while data quantity and data quality are of major importance for the successful implementation of AI methods. It is therefore essential that BD technologies are also considered during the research conducted on AI. In addition, the task includes the analysis of processes and concepts that enable the use of AI (e.g. data preparation or data mining). Especially the design of these processes must be intensified, if the applicability of AI technologies cannot yet be achieved due to adverse conditions. The research of the individual technologies, processes and IT structure is supported by identified literature in WP1 and is enhanced whenever necessary.
Interviews and a workshop session support the clustering towards application areas and sectors and enable a validation phase towards an individual technologies maturity level per sector. The Technology Readiness Level (TRL), which specifies the development of a technology on a scale from 1 to 9, supports the evaluation of the maturity of a technology within a sector. The resulting overview assesses the current state of the technology towards further research required and already being market ready (European Commission, 2017).
The TRL, industrial applicability scoring, and their implications on supply chains are assessed in order to evaluate the identified enabling technologies. In addition, the interviews and workshops serve to identify areas of collaboration between human operators with process control systems and plants and evaluates the respective acceptance level. The assessment dimension provides a backbone for the gap analyses in WP4, identifying technological gaps and implementation challenges as well as possible skill gaps with regard to human-machine-interaction.

T3.2 Mapping AI methods and BD technologies per sector and macro area (M 11-18)
(Lead IML, contribution ZLC, CNR, IRIS)
This work task develops a cross-comparison of the identified AI methods and BD technologies and the fields of application in different process industries (steel, chemicals, etc.). The approach enables a classification of the individual AI methods and BD technologies in the respective industries. The task aims at allocating AI methods and BD technologies with a certain degree of maturity to other industry sectors that have a similar field of application. The task results in concrete application possibilities developed individually for each sector. It furthermore develops an overview on possible and necessary combinations of AI methods and BD technology. The task also identifies possible barriers for the use of an AI methods and BD technologies in each industry sector along with the transfer of solutions between the sectors. Thus, drawing on these activities, task 3.2 provides an overview of the major challenges faced in the introduction of AI methods and BD methods and identifies areas in which overcoming these challenges result in an increasing economic value. The development and evaluation of the cross-sectoral application areas for AI methods and BD technologies will be done in close cooperation with the SPIRE and ICT community. The resulting map identifies potential for an increased exploitation of AI and BD approaches throughout the sectors of the process industry.

T3.3 Methods for the exploitation of AI methods and BD technologies (M 15-17)
(Lead IML, contribution IRIS)
This task identifies further enabling AI methods and BD technologies within areas wherein current approaches of the process industry are either not transferable or difficult to apply or implying to overcome a strong economic or technological gap. The task, hence, refers to other industries with similar requirements to identify possible AI methods and BD technologies to overcome this gap. With the support of a detailed literature research, the activities of T3.3

compile methods and approaches (e.g. data generation or data curation), which will enable an easier application of AI methods and BD technologies for the respective fields of the process industry. This task serves to identify strategies for the development of AI applications and the enhancement of BD technologies, inter alia focusing on the generation of data – a necessity for AI applications.

T3.4 Industry validation (M 16-19)
(Lead PNO, contribution all)
In task 3.4 "industry validation", the mapped methods and the maturity level of the individual technologies identified in the respective process industries are validated in cooperation with the SPIRE association. The methods and their maturity level as well as a possible cross-sectoral transfer are proposed to relevant stakeholders from previous WPs (T1.2 and T2.1). Those will be involved through a dedicated workshop project. Target groups, especially policy makers, companies, and researchers from the addressed sectors will actively discuss, integrate and provide suggestions and feedbacks. The SPIRE associations would serve as contact point for the validation, to address a wide range of users and, more in general, industry stakeholders. The results of this validation enable a realistic representation of the current and potential applications of AI and BD technologies in the process industry.

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D3.1	Outline on applied AI methods & BD technology per sector	3 - Fraunhofer	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D3.2	Cross-comparison between industries and possible areas of transfer	3 - Fraunhofer	Report	Confidential, only for members of the consortium (including the Commission Services)	17
D3.3	Overview on validated methods per sector and validated methods for cross-sectoral and crossindustrial transfer	1 - PNO	Report	Confidential, only for members of the consortium (including the Commission Services)	19

Description of deliverables

D3.1 Outline on applied AI methods and BD technology per sector (M18) (IML). Overview of current AI technologies with its corresponding TRL levels. The deliverable stresses identified gaps per sector and field of activity and highlights potential cross-sectoral areas of exchange.
D3.2 Cross-comparison between industries and possible areas of transfer (M17) (IML). List of potential tools and methods from other industries to overcome identified gaps and challenges in the application of AI methods and BD technologies.
D3.3 Overview on validated methods per sector and validated methods for cross-sectoral and cross-industrial transfer (M19) (PNO). Evaluation and validation of the developed application areas set maturity levels of AI methods and BD technologies in the process industry and evaluated potential of transfer between the sectors of the process industry with the help of industrial application partners. The result is the identification of a realistic potential and feasibility analysis.

D3.1 : Outline on applied AI methods & BD technology per sector [18]

Overview of current AI technologies with its corresponding TRL levels. The deliverable stresses identified gaps per sector and field of activity and highlights potential cross-sectoral areas of exchange.

D3.2 : Cross-comparison between industries and possible areas of transfer [17]

List of potential tools and methods from other industries to overcome identified gaps and challenges in the application of AI methods and BD technologies. D3.3 : Overview on validated methods per sector and validated methods for cross-sectoral and cross-industrial transfer [19] Evaluation and validation of the developed application areas set maturity levels of AI methods and BD technologies in the process industry and evaluated potential of transfer between the sectors of the process industry with the help of industrial application partners. The result is the identification of a realistic potential and feasibility analysis.

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS1	Map and Analysis of AI&BD technologies in industrial processes	4 - IRIS	18	AI&BD technologies mapped and analysed per industrial processes in SPIRE sectors
MS4	Mapping of AI&BD technologies and maturity level assessment	3 - Fraunhofer	18	D3.3 on 'Overview on validated methods per sector and validated methods for cross-sectoral and crossindustrial transfer ' submitted to the European Commission

Work package number [9]	WP4	**Lead beneficiary** [10]	2 - ZLC
Work package title	Roadmap and industry synergies		
Start month	12	**End month**	24

Objectives

Based on the previous work packages, WP4 will define a roadmap on AI and BD of application in process industries for all the SPIRE industrial sectors with specific recommendations to guide researchers, managers and operators in the implantation of these technologies. For this, a structured analysis of gaps and opportunities of AI and BD technologies (identified in WP3) and their current level of penetration will be performed. A study on the applications of these technologies and their potential will allow identifying knowledge transfer and synergies between sectors to be exploited. The obtained roadmap will be validated by industry and interested stakeholders, boosting knowledge sharing among different processes and sectors.

Description of work and role of partners

WP4 - Roadmap and industry synergies [Months: 12-24]
ZLC, PNO, Fraunhofer, IRIS, CNR
T4.1 Gaps and opportunities analysis – recommendations (M 15-18)
(Lead ZLC, contribution IML, CNR, PNO)
T4.1 focuses on the identification of gaps between the application of AI and BD in high-tech, tech-focused firms/ industries versus process industry to detect opportunities for collaboration and technology transfer as well as possible further synergies that may result from such collaboration or transfer among sectors. Such opportunities for the SPIRE sectors in development paths towards the transformed industrial sector will be coherently formulated.
Based on a close collaboration with WP2 and the mapping on AI and BD technologies per sector and macro area performed in T3.2, the potential industrial processes and collaborations that can benefit from these technologies are identified. This will help managers and researchers better understand the commonalities across different processes and technologies that make them suitable (or not) for AI adoption. The analysis of the current AI and BD technologies used by sector, identifying current processes and applications, will expose the opportunities and complications arising when leveraging synergies against the implementation of these technologies in other processes / activities across the sectors. Assessment of the feasibility of AI and BD technologies in the SPIRE sector and consultation with experts (workshops and interviews) for the identification of similar or complementary processes / activities where the AI or BD technology can be used. Examples of barriers/opportunities could be:
a. Opportunities: synergies and cooperation/coordination between different processes / activities for one actor. Synergiesand cooperation/coordination between different actors in the value chain
b. Barriers: regulatory, organizational, information exchange restrictions, financial, technologicalc.
Business models that enable the above opportunities
Furthermore, the analysis on gaps and opportunities can be extended, exploring whether specific gaps exist towards answering an industrial requirement and also whether specific collaborations or knowledge transfer could benefit the development of AI and BD technologies, exploit successful solutions for innovative technologies and recommendations. T4.1 will deliver a formulation of open challenges based on the examination AI and BD mapping technologies per sector and macro area, their application, level of penetration and potential.

T4.2 Case Studies and Business Models (M12-23)
(Lead ZLC, contribution all)
Together with T4.1 and using the output of WP2, Task 4.2 will perform an analysis of the current situation of the SPIRE industry, its implemented AI and BD technologies and degree of penetration, and the main existing business models in use, in order to highlight the opportunities and challenges regarding the implementation of new data driven technologies and the impact on new business models.

The input from previous tasks in WP4 would shed some light on whether it is the actual business model and/or the supply chain structure (e.g., production process, contracts, quality control, distribution ..) or the technologies available for the process industry that lead to the rather low level of adoption of AI and BD in comparison to technology focused firms. In certain cases, it could be both the traditional business models and supply chain structure that prevent firms in the process industry to implement AI and BD solutions.
In order to complement this analysis and make sure that the findings are practically relevant for different stakeholders for the process industry, a business model game similar to the one used in the INSPIRE project will be developed. Due

to the 'new normal' related to the current COVID-19 emergency, PNO will consider the possibility of virtual meeting to play the business model game online, in case physical meetings opportunities are restricted. In any case, a combination of face-to-face and online would potentially be a good implementation option, depending on the situations. In this game (simulation), each partner would select a real-life use case from a particular sector (e.g., steel, ceramics, etc.) concerning a problem/opportunity (e.g., maintenance to increase the uptime of resources) that the relevant stakeholders of this business case face, where AI and BD technologies might offer some solutions. Our goal in this task is to cover all eight sectors of the SPIRE community.
The goal is to organize workshops, to be held in person and/or online, for each use case, bringing relevant stakeholders together (e.g., manufacturer, supplier of raw materials, distributors, policy makers...). The workshops be run face-to-face or online, and the interaction will be moderated by experts from PNO and other AI-CUBE partners. The stakeholders will be encouraged to brainstorm about the challenges/barriers and the potential of the AI and BD technologies that could be used to overcome them, while they play the business model game.

Specifically, in this task, we will do the following based on the MAMCA methodology used in previous WPs:
• Develop the game by defining what potential "actors/roles" (e.g., technology providers, AI solution providers,distributors, manufacturers, policy makers, etc.) should be included in each use case,
• Define the value chain configuration (dynamics, relationships, infrastructure) and identifying the "influencers" and"followers" in regard to AI and BD solutions,
• Define the game rules facilitating the discussion among players in order to find out:
o The current (as-is) value chain configuration and processes,o The current (as-is) AI-related (or similar) technologies in use,
o What the main business problem/opportunity is,
o The available AI and BD technologies that might be available for use in this particular case with some potential tohelp solve the problem identified in the previous step,
o The available new business models (e.g., reconfigured supply chains, new contracts and optimized processes withinthe same configuration) with potential to help solve the problem identified in the previous step,
o The "to-be" business model/AI and BD solutions pair with the most potential and the challenges involved,
• Develop a post-game survey/discussion to summarize and consolidate findings
• Main conclusions on how to generate and exploit synergy between processes and industries based on data-driventechnology implementation and new business models.

In addition to the findings that come out of the workshops (games being played by the relevant stakeholders), the game itself, both in its soft and hard versions, will also be a deliverable of this task, which could be used in the future by other companies and sectors to bring all actors together and reach a workable solution in regard to the use AI and BD in process industry.

T4.3 Development of the roadmap per sector and area (M 17-24)
(Lead ZLC, contribution CNR, IML)
This task will gather all the results of T4.1 and translate into a roadmap tailored per sector and technology, that will:
• Propose novel and innovative business solutions that would help remove barriers and provide new opportunities forAI and BD technologies in new processes and sectors according to the different macro areas of application.
• Integrate the use of AI and BD technologies with current business models and organizational models.
• Develop simple tools to aid these sectors to perform cost-benefit analysis before making significant investment in AIand BD technologies.
• Change in management to eliminate barriers and reactionary behaviour (organizational, business relational, training).
• Pay particular attention to the requirements of SMEs to increase the competitiveness of the SPIRE sectors.
• Also, openly available strategies of European industry will be considered where applicable and relevant.

The AI-CUBE roadmap in AI and BD technologies will be a guideline in the implementation of these technologies allowing all the SPIRE sectors to benefit and share the knowledge gained to boost the competitiveness of the European industry.
AI-CUBE will implement the SRIA on Artificial Intelligence developed by the EC Communication as well as other European policies and guides elaborated by the EC and the AI HLEG.

T4.4. New paths of collaboration for ELSI (M 20-24)
(Lead CNR, contribution ZLC)
Revision of the ethical, legal and social implications of AI and BD developments, considering the values encoded in algorithms, the need to evaluate outcomes, and issues of bias and transferability, data ownership, confidentiality and consent, and legal, moral and professional responsibility. The potential effects for clients and different actors in

the supply chain (human operators), including confidentiality and process control systems and plants for seamless collaboration.

The ethical, legal and social analysis will consist of the following components:
State of the art review

- Analysis of legal and human rights requirements in and outside the EU
- Survey of Research Ethics Committee approaches and codes
- Ethical assessment
- Identification of the skills required to allow a seamless collaboration of human operators• Gap analysis
- Developing ethical framework and skills requirements

T4.5. Validation (M 20-24)
(Lead PNO, contribution all)
The resulting roadmap will be validated within a stakeholder workshop where SPIRE and other relevant stakeholders from the industrial sector will be invited to point out items that have not been considered and provide their input on feasibility and detailing of the proposed opportunities for collaboration, technology transfer and synergies.
Expert groups per sectors and technologies will be formed for the validation to ensure its successful deployment and application, involving different stakeholders such as researchers, managers, operators planning, technology providers, SMEs, etc., and allowing intercross contributions and add final conclusions.
The validation will consist of specific workshops per sector and technology and the exchange of the results with other sectors.

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D4.1	Business Model Game development	1 - PNO	Demonstrator	Public	18
D4.2	Gaps and opportunities analysis – recommendations and business models	2 - ZLC	Report	Public	18
D4.3	AI-CUBE roadmap	2 - ZLC	Report	Public	24
D4.4	New paths of collaboration for ELSI	5 - CNR	Report	Public	24

Description of deliverables

- D4.1 Business Model Game development (M18) (PNO): Development of a Business Model Game including thedesign of the tool strategy and rules.
- D4.2 Gaps and opportunities analysis – recommendations and business models (M18) (ZLC): Report analysis of thecurrent situation of the SPIRE industry, its implemented AI and BD technologies and degree of penetration, and the main existing business models in use, in order to highlight opportunities and challenges regarding the implementation of new data-driven technologies and the impact on new business models
- D4.3 AI-CUBE roadmap (M24) (ZLC): report gathering the results of D4.2 translated into a roadmap serving as aguideline in the implementation of AI and BD technologies tailored to all the SPIRE sectors.
- D4.4 New paths of collaboration for ELSI (M24) (CNR): Report on the revision of the ethical, legal and socialimplications of AI and BD developments.

D4.1 : Business Model Game development [18]

Development of a Business Model Game including the design of the tool strategy and rules.

D4.2 : Gaps and opportunities analysis – recommendations and business models [18]

Report analysis of the current situation of the SPIRE industry, its implemented AI and BD technologies and degree of penetration, and the main existing business models in use, in order to highlight opportunities and challenges regarding the implementation of new data-driven technologies and the impact on new business models D4.3 : AI-CUBE roadmap [24] Report gathering the results of D4.2 translated into a roadmap serving as a guideline in the implementation of AI and BD technologies tailored to all the SPIRE sectors D4.4 : New paths of collaboration for ELSI [24] Report on the revision of the ethical, legal and social implications of AI and BD developments.

Schedule of relevant Milestones				
Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS5	Validation of Roadmap and Guidelines for AI&BD technologies implementation in process industry	2 - ZLC	24	D4.2 on 'AI-CUBE roadmap' submitted to the European Commission

Work package number [9]	WP5	**Lead beneficiary** [10]	1 - PNO
Work package title	Dissemination, Communication and Exploitation		
Start month	1	**End month**	24

Objectives

The purpose of WP5 is to develop and implement a multitude of measures to communicate project findings and facilitate the industry engagement in their implementation. The goal is to maximize awareness around the project results and create a sustainable dissemination and exploitation platform, based on stakeholder involvement at regional, national and European levels.

Specific objectives of WP5 are:

- Creating the AI-CUBE dissemination and exploitation network that will enable EU industry to share knowledge andexpertise gained in this project on how to generate and exploit synergy of AI and BD deployment and penetration across all the 8 SPIRE process industries, based on a specific roadmap and guideline designed for the implementation of AI and BD technologies in the European process industry.
- Raising awareness and inform stakeholders about the AI-CUBE project, its objectives, and results, and the relevanceof its findings. The main effort will be devoted to communicating information about project objectives and scope and disseminating the results across the European process industry community and stakeholders.

Validating the AI-CUBE roadmap and guideline for AI&BD implementation in process industry in a contained set of stakeholders, identified through specific stakeholder and MAMCA methodologies, to guarantee a widely agreed consultation base in a European-wide network of stakeholders from cross-sectorial industrial parks.

Description of work and role of partners

WP5 - Dissemination, Communication and Exploitation [Months: 1-24]

PNO, ZLC, Fraunhofer, IRIS, CNR

Task 5.1 Definition of the strategy for dissemination of project results (M 1-3)

(Lead PNO)

The dissemination strategy and plan will be developed with the aim of achieving the maximum involvement of stakeholders in the different target groups, including:

- Definition of dissemination objectives and corresponding target groups;
- Identification of the dissemination instruments and events (e.g., conferences, magazines, web sites, onlinecommunities), including the specification of the AI-CUBE Project dissemination and exploitation platform;
- Definition of dissemination methodology detailing the content and timing of the various dissemination activities, tasksand responsibilities assigned to the different involved partners;
- Provision of structures for the dissemination and information actions/materials (structural design of the website,template, frequency and content of newsletters, standard lay out for presentations and reports).

Task 5.2 Communication activities (M 1-24)

(Lead PNO, contribution all)

Communication related to promoting the AI-CUBE project and its results will be implemented through strategic and targeted measures for communicating and possibly engaging in a two-way exchange with the targeted groups. Specific communication tools to be developed will be:

- Definition of a project visual identity for AI-CUBE, supported with a logo, leaflets, brochure, poster, roll-up, etc.- AI-CUBE website
- Periodic newsletter
- Social media campaigns
- Scientific communication of project results through the participation in international conferences, workshops,seminars, and the publication in scientific journals.

Task 5.3 Exploitation and dissemination (M1-24)

(Lead PNO, contribution all)

The AI-CUBE roadmap and guidelines will be validated in a contained set of stakeholders in three phases: inwardly oriented, and externally oriented, using good practices and tools from European-wide network of cross-sectorial industrial community and the experiences of the SPIRE network.

At least one video will be realized to provide an easy to grasp dissemination of the results, along with infographics, presenting the CUBEs resulting from the analyses conducted in WPs 3-4, visualising the current status and penetration

of AI and BD in different process industries and core processes; the AI-CUBE AI and BD Business Model Game, to help industries design in their AI and BD business model of the future; eventually inspirational future AI and BD Business Cases, to stimulate thinking ant take up; the designed roadmap, including detailed descriptions on how AI and BD could be better developed and taken up.

The Dissemination, communication & Exploitation Work Package will be carried out under the responsibility of PNO. Further details are outlined in section 2.2.

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D5.1	Dissemination Plan	1 - PNO	Report	Public	3
D5.2	Website and AI-CUBE dissemination and exploitation platform	1 - PNO	Websites, patents filling, etc.	Public	4
D5.3	A - Report on Dissemination, Communication & Exploitation activities	1 - PNO	Report	Public	12
D5.4	B - Report on Dissemination, Communication & Exploitation activities	1 - PNO	Report	Public	24

Description of deliverables

- D5.1 Dissemination Plan (M3) (PNO): Report on the dissemination strategy and plan.
- D5.2 Website and AI-CUBE dissemination and exploitation platform (M4) (PNO): Online launch of the website andAI-CUBE dissemination and exploitation platform
- D5.3. A. Report on Dissemination, Communication & Exploitation activities (M12) (PNO): Report on AI-Cubedissemination, communication and exploitation and reports the related activities conducted until month 12. - D5.3. B. Report on Dissemination, Communication & Exploitation activities (M24) (PNO): Report on AI-Cube dissemination, communication and exploitation and reports the related activities conducted until month 24.

D5.1 : Dissemination Plan [3]

Report on the dissemination strategy and plan.

D5.2 : Website and AI-CUBE dissemination and exploitation platform [4]

Online launch of the website and AI-CUBE dissemination and exploitation platform

D5.3 : A - Report on Dissemination, Communication & Exploitation activities [12]

Report on AI-Cube dissemination, communication and exploitation and reports the related activities conducted until month 12.

D5.4 : B - Report on Dissemination, Communication & Exploitation activities [24]

Report on AI-Cube dissemination, communication and exploitation and reports the related activities conducted until month 24.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS6	Dissemination plan	1 - PNO	3	D5.1 'Dissemination Plan' submitted to the European Commission
MS7	Website and AI-CUBE dissemination and exploitation platform running	1 - PNO	4	Open and public access to the Website and AICUBE dissemination and exploitation platform

Work package number [9]	WP6	**Lead beneficiary** [10]	1 - PNO
Work package title	Project coordination and quality management		
Start month	1	**End month**	24

Objectives

The goal of the WP6 on Project coordination and quality management is to provide the Consortium of the AICUBE Project with efficient and effective management and administration of the internal and external activities and relationships.
In pursuing this goal, the consortium will design an adequate management structure of the project to handle the coordination of the work carried out in the AI-CUBE Project, quality control activities, financial and thematic reporting, and all the communications among project partners (PPs), associated members, interested stakeholders, and the EC.
Specific objectives are:
- Managing the communication in the Project Consortium and Decision-making board
- Providing an overall project governance and quality management strategies ensuring the timely achievement of projectgoals as identified in the work plan
- Defining a central contact point to the project and managing efficient interface with the EC and other partners
- Ensuring coherence with all legal and EC requirements, rules and regulation, and supervise the compliance to theConsortium Agreement, including financial and legal management
- Managing the "AI-CUBE Expert Group" and the "AI-CUBE Network on AI & BD implementation in the EU Process Industry"

The Project Coordination and Quality Management Work Package will be carried out under the responsibility of the coordinator PNO. Further details are outlined in section 3.2.

Description of work and role of partners

WP6 - Project coordination and quality management [Months: 1-24]
PNO, ZLC
Task 6.1. Consortium and project administrative management (M 1-24)
(Lead PNO)
- Ensuring correct distribution and accountability for the funds received by the EC
- Establishing functional management structures for fruitful interactions among partners, efficient communication andreporting to the EC (e.g. implementing project management, work plan, financial and activity reporting tools)
- Aiding and advising PPs on administration and reporting tasks development
- Monitoring budget allocation and expenditure as well as effective and efficient execution of contract and work plans
- Representing the AI-CUBE Consortium in the interaction with the EC and other institutions
- Organising regular meetings of the AI-CUBE Project Consortium and Decision-Making Board
- Elaborating periodic Management and Progress Reports to regularly inform the EC on the activities of the Project(work progress, financial statements, etc.)

Task 6.2. Monitor technical Developments and Work Plan (M 1-24)
(Lead PNO)
- Managing interrelations between the different work packages of the project
- Monitoring technical advances of the project in each WP through a close interaction with all the WP leaders
- Ensuring advancement of the project according to the work plan defined in this proposal
- Observing external developments and act accordingly for strategic adaptation of the project activities
- Coordinating the establishment of the "AI-CUBE Expert Group" and the "AI-CUBE Network on AI & BDimplementation in the EU Process Industry" and maintain with them a close relation to validate the AI-CUBE activities and results.

Task 6.3. Quality Management (M 1-24)
(Lead PNO)
The quality management of the AI-CUBE Project entails a continuous check of the results and achievement of each WP, and the assurance that they are aligned with the objectives outlined in the work plan. The goal is to ensure that the project orientations, activities, and outcome are properly answering the needs of the EU process industry of efficiently

applying AI and BD technologies. Moreover, task 6.3 aims at securing a timely completion of the deliverables according to the work plan and provide quality information.
The general processes for quality management, including the definition of a quality assurance plan, are described in further details in section 3.2.2.
Within Task 6.3., these processes will be reviewed and updated as needed within the first weeks of the AI-CUBE Project, in order to adapt to the evolving workflow.
The task leader, PNO, will be responsible for the observation of the review processes and proposition of actions to face undesirable/unexpected developments. These Consortium will then discuss these processes and PNO will secure the follow up to the decisions taken. All PPs will be incorporated in the reviewing and monitoring workflow for the WPs of their concern. WP leaders will play a significant role in quality management activities.
Task 6.4. Data Management (M 1-24)
(Lead PNO)
As to comply with the requirement of the Open Research Data Pilot (ORDP), the AI-CUBE consortium will draft a Data Management Plan at M6 including information about "findable, accessible, interoperable and re-usable" data, known as FAIR data. The DMP will provide information about: the handling of research data during the AI-CUBE project and after its conclusion; which data will be collected, processed and generated; which methodologies and standards will be applied; whether the data will be shared or made open access and how they will be curated and store, even after the conclusion of the project.

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D6.1	Project Management Plan	1 - PNO	Report	Public	2
D6.2	Quality Management Plan	1 - PNO	Report	Public	2
D6.3	Data management Plan – DMP	2 - ZLC	ORDP: Open Research Data Pilot	Public	6

Description of deliverables

- D6.1. Project Management Plan (M2) (PNO): Project Management Plan
- D6.2. Quality Management Plan (M2): (PNO): Quality Management Plan
- D6.3 Data management Plan – DMP (M6): (ZLC) Data Management Plan

Other deliverables such as periodic management reports are mandatory and contractually foreseen, therefore they are not listed here.

D6.1 : Project Management Plan [2]

Project Management Plan

D6.2 : Quality Management Plan [2]

Quality Management Plan

D6.3 : Data management Plan – DMP [6]

Data management Plan

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS8	Kick-off meeting	1 - PNO	1	Kick-off meeting executed successfully

MS9	Quality Management procedures implemented	1 - PNO	24	Compliance of all partners to deliverable D6.2 'Quality Management Plan'

1.3.4. WT4 List of milestones

Milestone number[18]	Milestone title	WP number[9]	Lead beneficiary	Due Date (in months)[17]	Means of verification
MS1	Map and Analysis of AI&BD technologies in industrial processes	WP1, WP2, WP3	4 - IRIS	18	AI&BD technologies mapped and analysed per industrial processes in SPIRE sectors
MS2	Consultation with stakeholders	WP2	1 - PNO	9	Stakeholders engagement
MS3	Assessment framework defined	WP2	5 - CNR	9	Definition of the criteria to describe and assess the AI&BD technology level of penetration and maturity.
MS4	Mapping of AI&BD technologies and maturity level assessment	WP3	3 - Fraunhofer	18	D3.3 on 'Overview on validated methods per sector and validated methods for cross-sectoral and crossindustrial transfer ' submitted to the European Commission
MS5	Validation of Roadmap and Guidelines for AI&BD technologies implementation in process industry	WP4	2 - ZLC	24	D4.2 on ‘AI-CUBE roadmap’ submitted to the European Commission
MS6	Dissemination plan	WP5	1 - PNO	3	D5.1 'Dissemination Plan' submitted to the European Commission
MS7	Website and AI-CUBE dissemination and exploitation platform running	WP5	1 - PNO	4	Open and public access to the Website and AICUBE dissemination and exploitation platform
MS8	Kick-off meeting	WP6	1 - PNO	1	Kick-off meeting executed successfully
MS9	Quality Management procedures implemented	WP6	1 - PNO	24	Compliance of all partners to deliverable D6.2 'Quality Management Plan'

1.3.5. WT5 Critical Implementation risks and mitigation actions

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
1	Budget is not spent correctly, or contributors do not deliver according to the plan (Low)	WP6	For these cases the Consortium Agreement will provide tools for the coordinator and the consortium to react timely and redistribute the work and/or budget. Deviations from the work plan and budget will be identified on short term through the work in Task 6.2. Furthermore, frequent meetings will enable the consortium to make efficient use of these tools when needed.
2	Delay propagation from technical work packages (Medium)	WP1, WP2, WP3, WP4, WP5, WP6	Work has to be stringently and efficiently managed, planned and monitored. This will be done e.g. through a close follow-up by the coordinator and the review meetings with the work package leaders. The compliance has to be followed up through quality management.
3	The needs of stakeholders of process industry and AI&BD sector are not properly addressed within the project (Low)	WP2, WP3, WP4	The Expert Group will represent a larger community within the process industries and AI and BD sector. The Experts will be specifically requested to provide feedback on behalf of all the actors represented (research institutions, large corporations, SMEs, etc.). AI-CUBE will involve the SPIRE association and other relevant ones to ensure the involvement of all the interested stakeholders.
4	Stakeholder network does not represent all relevant stakeholders. (Low)	WP4, WP5, WP6	The project consortium is very well-connected with a large part of the relevant stakeholder network. The completeness of the stakeholder network will be supported through the Expert Group that will be connected to the project.
5	Stakeholders will not reveal information due to IPR issues. (Low)	WP5, WP6	All considerations should regard pre-competitive developments and data confidentiality will be taken into consideration.
6	Relevant experts and stakeholders will not be able to participate in consulting and validation processes. (Medium)	WP1, WP4, WP5, WP6	Relevant experts and stakeholders will be able to closely follow up all processes and results through the interactive website and dissemination and exploitation platform (WP5). Those stakeholders will be contacted directly and supported to participate through online means. Furthermore, the coordinator will plan and inform the experts in a timely manner about the consulting processes and therewith the experts can plan the activities.

1.3.7. WT7 Tentative schedule of project reviews

Review number [19]	**Tentative timing**	**Planned venue of review**	**Comments, if any**
RV1	12	tbd	12 month after start of CSA
RV2	24	tbd	end of CSA

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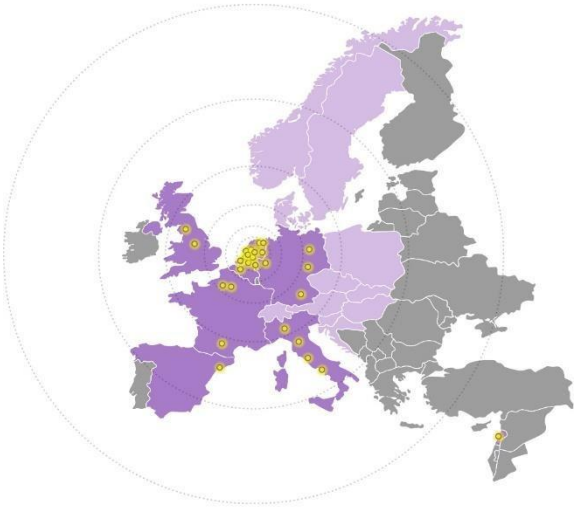
4. Section 4: Members of the consortium

4.1. Participants (applicants)

Partner	**CiaoTech s.r.l / PNO Group B.V.**		
Short Name	PNO	**Type**	Large Enterprise
Role	Coordinator and Innovation Management services	**Country**	Italy
Website	www.pnoconsultants.com	**Logo**	PNO CiaoTech

Description of the legal entity



CIAOTECH s.r.l., a company 100% owned by the PNO Group B.V. (hereafter PNO), is the largest independent public funding and innovation consultancy in Europe, with 30 years of hands-on expertise with more than 600 innovation projects funded per year in most EU countries, annually raising approximately 1 Billion Euro for its clients. Created in 1985, PNO is a high-growth knowledge-intensive company, supporting over 2.000 clients throughout Europe, annually developing over 250 European consortia. PNO has offices in 7 European countries and in Israel, employing in around 500 professionals including scientists, engineers, consultants, a Brussels policy advisory service, as well as financial and legal experts, with consolidated experience in innovation processes and funding in international working environments. The company specialised in Grants and Innovation Management, providing support services in Innovation processes, Technology Transfer, IT solutions and funding for research, development and innovation. PNO "Innovation Management" services deliver high quality support to large companies,
SMEs, Universities, Research Institutes, Associations and clusters in the full cycle of the innovation process, including:

- Analysing, defining and planning innovation processes; • Building innovation networks, partnerships and projects;
- Managing Projects and driving innovation.

PNO projects involve both small, medium sized and multinational industries, as well as Universities and Research institutes, amongst which are European and international leaders such as Amsterdam Schiphol Airport, BASF, Booking.com, DOW Chemical, CERN, Finmeccanica/Selex, Fraunhofer Institute, Harbours of Rotterdam and Antwerp, HP, Intel, Microsoft, Philips, Procter & Gamble, Solvay, TomTom, the Universities of Bordeaux, Ghent, Leuven, Rome, Naples, Utrecht, Warwick, Politecnico of Milano, Turin and many more.
PNO is key partner and advisor to the European Technology Platform (ETP) SusChem and the European Council for the Chemical Industry (CEFIC) and PNO recently (2019) co-developed the latest Strategic Innovation and Research Agenda. PNO has furthermore supported A.SPIRE in the development of its first Roadmap and recently (2019) the annual monitoring report on SPIRE funded projects . Furthermore, PNO manages several European EC funded network projects to build communities of valuable SMEs, large industries, Universities and Research Institutes, to further innovation in specific sectors, such as the bio-based product market (BIOCHEM), transport (SMART), security (OSMOSIS), open source and embedded software (SHARE), Chemical Regions for Resource Efficiency (R4R) to name a few.
Leveraging on its insights and knowledge on innovation methodologies and practices, PNO increasingly supports its clients as partner in strategic European research, development and innovation projects, to carry-out market and

functional analyses, develop and implement road mapping, dissemination and exploitation strategies and business plans and support professional project management.
PNO participates in a significant number of European H2020 project that relate to the topic of the AI-CUBE one. Specifically, PNO has recently concluded the INSPIRE Coordination and Support Action (CSA) targeted at analysing the future of the European process industry, more particularly how novel technologies, such as AI, may lead to novel Business Models for the eight SPIRE process industries. The Business Model Innovation Game developed together with SPIRE partner TNO, as well as the research needs paper from INSPIRE, will provide the basis for a dedicated AI-CUBE Business Model Game which can help to focus the AI and BD roadmap developments. Furthermore, PNO participated as partner to the NEXT-NET Coordination and Support Action, targeted at analysing long-term megatrend-scenarios for the process – manufacturing – logistics supply chain, and developing a research roadmap for the supply chain of the future. The output of INSPIRE and NEXT NET will provide direct knowledge and tool base for the AI-CUBE project.

Main tasks and responsibilities within the project

PNO will be the coordinator of the project and therefore lead WP6. Also, PNO will lead WP5 "Dissemination, Communication and Exploitation", that is designed to maximize awareness around the project results and create a sustainable dissemination and exploitation platform, based on stakeholder involvement at regional, national and European levels. Also, PNO will technically contribute to the other work packages with their expertise.

Profiles of persons primarily responsible for carrying out the proposed activities

Ronald Weerdmeester	**Profile**	https://www.linkedin.com/in/ronweerdmeester/

Several decades of consulting experience in Innovation strategy, visions, roadmaps and strategic research and innovation agendas development.

Ron WEERDMEESTER currently works as PNO Group Innovation Manager. Graduated in law at Rijksuniversiteit Groningen and holds a Certificate Degree in "Information Systems in Organisations", from Boston University. Member of the board of the PNO's units PNO Innovation and CiaoTech. Expert in innovation strategy, intelligence and innovation funding for industries, academia and public organisations. 25 years' experience in development of technology development projects with a European dimension and EU funding (from FP4 to HORIZON 2020). Development of more than 100 successful EU projects. Co-author of multiple visions, roadmaps and strategic research and innovation agendas for sector organisations (process industry, chemistry, bioeconomy, water). Project manager of various Economic and Technological Intelligence actions, assisting industry in their innovation needs, developing European networks. Regular training to SMEs, multinationals and research organisations from various European countries, on issues as European research and innovation policies, open-innovation project building, project management in networks, technology transfer and the development of European partnerships.

Chiara Eleonora De Marco	**Profile**	www.linkedin.com/in/chiaraedemarco

Chiara (female) is an Innovation Consultant in the Strategic Innovation Services Team of CiaoTech/PNO Group. She holds an International PhD in Innovation Management from the Scuola Superiore Sant'Anna, Pisa (IT), a Master's Degree in International Relations and International Protection of Human Rights, and another one in Diplomatic Studies, both awarded form the Società Italiana per l'Organizzazione Internazionale (SIOI) of Rome (IT). She graduated summa cum laude in International Law from LUMSA University, Rome (IT).
Chiara has working experiences both in National and International environments, embracing academia, public and private sector. She was Postdoctoral Scholar at Haas School of Business of the University of California, Berkeley (US), working on Corporate Capability Building Programs for Digital Transformation, and at the Institute of Management of Scuola Superiore Sant'Anna (Pisa, IT), conducting research and leading national and international projects on open innovation, digital business model and strategy, innovation policy, small business R&D, research spinoffs, technology and knowledge transfer. Over her years in Academia, she conducted research on and led consultancy project for the private sector (both SMEs and large corporations) and the public one (European Commission, JRC), joining theory and practice of open innovation to identify and help addressing bottlenecks and pain points in corporate strategy, but also recommendation for decision makers in the field of innovation.
As member of the Organizing Committee of the World Open Innovation Conference (https://woic.corporateinnovation.berkeley.edu/), she is in charge with the management of Industry Sessions and Workshops.

Chiara disseminated her research work in several international conferences and forums, joining the international innovation community across the EU and the US. Her work has been published on peer-reviewed, international academic journals, EU publications, national publications, specialised blogs and newspaper.

List of up to 5 relevant publications, and/or products, services, or other achievements relevant to the call content

Analysing, defining and planning innovation processes

- Market, stakeholder & Technology Intelligence. Provides a helicopter view on who's doing what in the EU and beyond, benchmarking target technology with other technologies and solutions, analysing and identifying the essential markets that need to be targeted;
- Systematic Innovation using advanced methodologies (eg: TRIZ). Discovers new technologies, partners and markets, structuring and driving open innovation processes in multi-stakeholder contexts, or for individual clients;
- Vision and roadmap development. Shows where to focus, organising, supporting and reporting on vision and innovation roadmaps of interest.

Building innovation networks, partnerships and projects

- Develop innovation projects, proposals and consortia. Builds innovation project with international partners supported by public funding, bringing the best funding expertise, relevant networks and hands-on support;
- Effective linking to innovation ideas, partners and funding across sectors and Europe. Enables becoming a partner in projects supported by public funding to discover or bring innovation and technology to new markets, unlocking networks of projects and stakeholders.

Relevant Publications

- The Value of Water, Multiple Waters, for multiple purposes and users, Ron Weerdmeester et al. (2016) – WssTP, The European Water Platform, URL: http://wsstp.eu/publications/
- Spire Roadmap, Ron Weerdmeester, Pablo Tello (2016) – SPIRE Sustainable Process Industry through Resource and Eneregy Efficiency, URL: www.spire2030.eu/what/walking-thespire-roadmap/spire-Roadmap
- Knowledge Management Maturity Model / Roadmap, Ron Weerdmeester, Chiara Pocaterra, Mark Hefke (2003) – Information Societies Technology Programme, URL: www.providersedge.com/docs/km_articles/KM_Maturity_Model_of_VISION_Project.pdf
- Supporting Technology Innovation Processes in Manufacturing, Paolo Salvatore et al. (2016) – 2nd International Conference on Quality Engineering and Management, Cluj-Napoca, Romania. URL: www.market-itproject.eu/wpcontent/uploads/2012/08/SMEs_Innovation_QIEM_2012-Final.pdf

List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

PNO's excellence and expertise on topics related to the main tasks in AI-Cube project is grounded on the involvement in several European projects in the area of supply chain, digital technologies as well as roadmapping and forecasting. Being partner in multiple EU-funded projects (FP7, H2020, SPIRE and eco-innovation projects) and several commercial assignments, PNO fully qualifies as a relevant and positive long-term stakeholder within the European process industry community. PNO has been the architect and is currently participating in several industry support networks with Support from the European Commission in the Security, Transport, Biotech and Chemical sectors, promoting the establishment of industry collaborations, knowledge transfer and technology acceleration. PNO is also involved as advisor in roadmap development for the chemical industry towards Public Private Partnership Programmes in HORIZON 2020 (eg: SPIRE), with an important focus on industrial processes, as well as European networking and incubator projects with various smart specialisation regions which will provide a real life experience base. In the SPIRE domain, the following list represents the most recent projects related to AI-Cube topics both from the technological and the organizational point of view:

2018-2022 DESTINY (IA - H2020 – SPIRE: Development of an Efficient Microwave System for material Transformation in Energy Intensive processes for an improved Yeld). The project focused on the development of an innovative, scalable, sustainable and resource efficient solutions for energy-intensive process industries. Within DESTINY, PNO is responsible for conducting strategic analysis and intelligence on

the market and stakeholder landscape, to develop innovative business models and strategies for the commercialization and impactful exploitation of the results generated by the project activities.

2017-2019: NEXT-NET (NMPB37-2016-CSA: Next generation technologies for networked Europe). The project developed a joint roadmap where the effort of different initiatives like EFFRA, SPIRE, ALICE and ICT initiatives were put together to define the future of supply chain based on a distributed model for customized and sustainable production and distribution. Starting from the creation of a European network involving the most important ETPs and PPPs, NEXT-NET reached a critical mass of relevant stakeholders from all targeted industries and scientific domains by mean of direct actions along the whole project life. The project identified future industrial scenarios for 2030, through the analysis of the socioeconomic analysis of trends and drivers in the market and their impact on innovation & technology development.

Furthermore, simulation or system dynamics were used to formalize the industrial scenarios. PNO's role in the project was to represent the interests of process industry, and keep the contacts with stakeholders, especially in the steel and chemical. PNO was in charge of the collaboration mechanisms with stakeholders and the validation of the action plan. Being the Project coordinator of the project INSPIRE, PNO kept the contacts with this project and assured the transferability of the results to optimize its impact on NEXT-NET. Moreover, PNO will maintained the contacts with the SPIRE community.

2016-2017: INSPIRE (CSA – GA n° 723748, SPIRE-06-2016-CSA: Business models for flexible and delocalized approaches for intensified processing). PNO and the AI-CUBE consortium will build on and follow up to their experience in the INSPIRE project, of which PNO was coordinator, while ZLC and CNR were partner. INSPIRE targeted at analysing market trends and their impact on supply chain configurations. INSPIRE explored how these configurations in industrial locations can be flexibly designed based on state of the art of enabling technologies, as well as how resource efficiency benefits and hence environmental footprint can be optimised. INSPIRE contributed to the analysis of the current European networks, for the first time supporting the integration between process industry and discrete manufacturing, and identifying current practices of delocalization, opportunities for and barriers against flexibility/delocalization, and current practices for joint resource management and industrial symbiosis. The role of PNO in the INSPIRE project concerns activities of networking in the European Technology Platform and PPPs, and with relevant stakeholders. PNO will build on this experience to conduct its activities in the AI-CUBE project.

2015-2020: MEMERE (RIA – GA n° 679933, H2020-SPIRE-05-2015: MEthane activation via integrated MEmbrane Reactors). The key objective of the MEMERE project was to the design, scale-up and validate a novel membrane reactor for the direct conversion of methane into ethylene with integrated air separation. The focus of the project was on allowing integration of different process steps in a single multifunctional unit and achieving significantly higher yields in comparison with the conventional reactor technologies, combined with improved energy efficiency. PNO role in the project was to develop business model and business plan; support exploitation plan and activities, and the dissemination, with special focus on industrial stakeholders.

2015-2019: REE4EU (IA, GA n° 680507, H2020-SPIRE-07-2015: Integrated High Temperature Electrolysis (HTE) and Ion Liquid Extraction (ILE) for a strong and independent European Rare Earth Elements (REE) Supply Chain). The project developed new technologies to recover and recycle Critical Raw Materials. The consortium was led by SINTEF, and PNO supported stakeholder analyses and business modelling. The REE4EU pilot has successfully treated several tons of in-process wastes and end-of-life products containing rare earth elements, resulting in the recovery of almost hundred kilos of rare earth alloys directly from the mixed rare earth oxides feedstock produced. The alloy will be reused in the manufacturing of permanent magnet products.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

PNO will make available to the project its virtual Open Innovation Platform www.innovationplace.eu, proprietary online networking, community building, management and dissemination tool, hosting more than 15.000 registered users, all with a relevant innovation profile.

Partner	Zaragoza Logistics Center		
Short Name	ZLC	Type	Research Center
Role	Partner	Country	Spain
Website	www.zlc.edu.es	Logo	ZLC

Description of the legal entity

Zaragoza Logistics Center (ZLC) is an international center of excellence for research and education in logistics and Supply Chain Management that actively engages with industry and the public sector to develop and disseminate knowledge to accomplish its mission. ZLC acts as a catalyst for innovation and research in the area of logistics and SCM. A core purpose of ZLC is to integrate supply chain management into the business enterprise.

ZLC was established by the Government of Aragon in Spain, in partnership with the MIT Center for Transportation and Logistics (CTL-MIT) and the University of Zaragoza. ZLC is directly linked to the development of PLAZA – the largest logistics park in Europe - and has partnered with the Center for Transportation and Logistics at MIT to form the MIT-Zaragoza International Logistics Program. Also, the MIT Global SCALE (Supply Chain and Logistics Excellence) Network was initially formed in 2004, with the opening of ZLC, and it now spans four continents with the addition of the Center for Latin American Logistics Innovation (CLI-Colombia), the Malaysia Institute for Supply Chain Innovation (MISI), the Luxembourg Center of Logistics (LCL) and the Ningbo Supply Chain Innovation Institute China (NSIIC). In 2006 the Spanish Ministry of Education and Science designated ZLC as the National Centre of Excellence for research in the area of logistics and supply chain management (CNC-LOGISTICA). CNCLOGISTICA acts as coordinator of the Spanish Technology Platform in Logistics, Intermodality and Mobility (Logistop).

The research program is aligned to Zaragoza Logistics Center's (ZLC) mission of focusing on enhancing economic growth and competitiveness through innovation. To this end ZLC has developed 100+ R&D collaborative projects with regional, national and European industry focused on logistics and SCM. At European level, ZLC has participated in 25+ European Union funded projects, being ranked as the Spanish entity with most EU funded projects in the field of logistics.

The Advanced Inventory Management Research Group works on issues focused on developing inventory management strategies and models to increase supply chain value, particularly using stochastic inventory models where randomness could originate from either demand or supply.

ZLC is also a founding member of ALICE (Alliance for Logistics Innovation through Collaboration in Europe), which is the European Technology Platform for Logistics. The Director, Dr. Susana Val Blasco, is a member of the ALICE Steering Group and ZLC faculty participate actively in ALICE Working Groups, one of them devoted to Systems & Technologies for Interconnected Logistics.

Furthermore, ZLC has a strong network of contacts and collaborators that involves relevant European and international contacts in the value chain sector, including policy makers, process industries, academia and research organisations.

Main tasks and responsibilities within the project

ZLC will lead WP4 devoted to the definition of a roadmap on AI and BD technologies of application in process industries for all the SPIRE industrial sectors. ZLC will analyse the gaps and opportunities of data driven technologies among processes and sectors looking for synergies. A cases study on business models will be performed following business models game developed in INSPIRE project.

ZLC's wide experience in the participation in European projects related to logistics and supply chain makes it the ideal partner to this consortium within the framework of the AI-CUBE project and participate in its technical workpackages.

Profiles of persons primarily responsible for carrying out the proposed activities

Mustafa Çagri	Profile	www.zlc.edu.es › dr-mustafa-cagri-gurbuz

Dr. Mustafa Çagri Gürbüz (male) is Professor of Supply Chain Management at the MIT-Zaragoza International Logistics Program. He is also a Research Affiliate at the MIT Center for Transportation and Logistics. He completed his PhD at the Department of Management Science at the University of Washington in July 2006. He earned his BS and MS degrees from the Department of Industrial Engineering at Bilkent University, Turkey in 1999 and 2001, respectively. His main research interests are inventory and supply chain management, optimizing distribution systems, contracts, and modelling operations systems. Mr. Gurbuz has been a consultant to AT&T Wireless, AREX, and RENFE. He has been a visiting professor at MIT CTL in 2009 and City University of Hong Kong in 2014. He has also published in journals such as Management Science and Operations Research Letters.

List of up to 5 relevant publications, and/or products, services, or other achievements relevant to the call content

- Boyonas M. I., Olavarría L., Sáenz M.J., Scenario Planning: Preparing the Future of Global Supply Chains. Designing and Implementing Global Supply Chain Management. IGI Global Book. Book Chapter number 4. Pp.51-91, December 2015 (Editors: Sudhanshu Joshi & Rohit Joshi).
- Val S., Kehinde A., Machado A., Ciprés C., Supply Chain Network Design. Tackling Regulations, Lead Time and Cost Efficiency. Sustainable Logistics and Supply Chains: Innovations and Integral Approaches (Contributions to Management Science). Springer, Book chapter, October 2015 (Editors: Joost de Bock & Meng Lu).
- Maestrini, V , Luzzini, D, Maccarrone, P., Caniato, F. Supply chain performance measurement systems: A systematic review and research agenda. International Journal of Production Economics. Vol. 183, Jan 2017, pp. 299-315
- Diaz, R., Marsillac, E. Evaluating Strategic Remanufacturing Supply Chains Decisions. International Journal of Production Research. Volume 55 (9), 2017, pp. 2522-2539
- Xu, J. Serrano, A. Lin, B., Optimal Production and Rationing Policy of Two-Stage Tandem Production System, International Journal of Production Economics, Vol. 185, March 2017, pp 100 – 112

List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

- **INSPIRE**: Towards growth for business by flexible processing in customer-driven value chains (01/09/201931/08/2018) - https://www.zlc.edu.es/research/projects/towards-growth-for-business-by-flexible-processing-incustomer-driven-value-chains/
- **NEXT-NET**: Next generation Technologies for networked Europe (01/10/2017 – 31/12/2019) - https://www.zlc.edu.es/research/projects/next-generation-technologies-for-networked-europe/
- **SELIS:** Towards a Shared European Logistics Intelligent Information Space – (01/09/2016 – 31/08/2019) - https://www.selisproject.eu/default.aspx
- **CLUSTERS 2.0:** Open network of hyper connected logistics clusters towards Physical Internet – (01/05/2017 – 30/04/2020) - http://www.clusters20.eu/
- **SENSE**: Accelerating the Path Towards Physical Internet - (01/10/2017 – 31/03/2020) - https://cordis.europa.eu/project/rcn/212876/factsheet/en

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Not expected

Partner	**Fraunhofer Gesellschaft zur Foerderung der angewandten Forschung e.v.**		
Short Name	Fraunhofer	**Type**	NGO
Role	Partner	**Country**	Germany

Website	www.fraunhofer.de/en.html	Logo	

Description of the legal entity

The Fraunhofer Society is the largest research organization for applied research in Europe. Its research fields are orientated towards persons' requirements: health, safety, communication, mobility, energy and the environment. The Society is spread amongst 60 institutes on different fields of applied science. Within AI-CUBE the Institute for Material Flow and Logistics (IML) participates, which is said to be the first address in holistic logistics research and currently combines digitalisation and artificial intelligence in their *Silicon Economy* strategy.
Fraunhofer IML was founded in 1981 in Dortmund, Germany, the IML currently occupies 315 employees as well as 250 post-graduates and students to works in all fields of internal and external logistics. In line with the Fraunhofer idea a twofold approach is followed: Fraunhofer IML once develops problem solutions for direct use by companies, and secondly, conducts preliminary research of two to five years. Teams form according to project and customer requirements to create cross-industry and customer-specific solutions, inter alia in the field of supply chain management & its simulation, closed loop economy and business process modelling. Artificial intelligence, smart finance and the Internet of Things are among the current research focal points. Fraunhofer IML coordinates the Internet of Things throughout Fraunhofer. With the current strategic focus on Silicon Economy, it explores new business models via platform economy approaches that centre on digitalisation and artificial intelligence.

Main tasks and responsibilities within the project

The key role from Fraunhofer IML within AI-CUBE lies within the expertise of AI and its use within supply chains and supply chain management. Fraunhofer IML supports the project twofold. Firstly, it provides strong experience in material stock management and supply chain for the process industry. Secondly, it supports on the use and application areas of AI within the field of supply chain management.
Fraunhofer IML will be work package leader of WP3. With its expertise for AI and knowledge about the process industry, Fraunhofer IML guides the categorization of current AI and BD technologies. Being involved in the EU project "Transforming Transport", Fraunhofer IML can refer to vast experience in the field of big data technologies in the area of transportation and create respective synergies. Its experience from the EU project "NextNet" supports methodologically for the evaluation of technologies used per sector and their level of maturity. Within the consortium Fraunhofer supports the exploitation of AI methods and BD technologies.
Fraunhofer IML furthermore involved in the tasks T1.3, T2.2 and T2.3, as well as 4.3 and 5.3 to enable insights towards AI throughout the project. It thus enables a continuous and streamlined approach and knowledge exchange.

Profiles of persons primarily responsible for carrying out the proposed activities

Saskia Sardesai	Profile	www.iml.fraunhofer.de/en/fields_of_activity/enterpriselogistics/supply_chain_engineering/team.html

Saskia Sardesai (female) spend her first working experience with a logistics service provider in India. Having joined Fraunhofer IML, she now focuses on supply chain planning, AI in supply chain management and simulation studies. In her current position as a Senior Scientist SCM Research and Strategy, she puts a strong focus on Research Projects with innovative Supply Chain and Collaboration aspects based on AI. She guided the EU CSA "Next Generation Technologies for Networked Europe" as a project leader from Fraunhofer IML.

Andreas Nettsträter	Profile	de.linkedin.com/in/andreas-nettstr%C3%A4ter-9894b086

Andreas Nettsträter (male) studied Computer Science, with minor Mechanical Engineering at the University of Dortmund. He is responsible for strategic initiatives with focus on European activities at the Fraunhofer-Institute for Material Flow and Logistics in Dortmund. His focus is on innovative IT solutions for logistics, supply chain management and manufacturing. He is also Executive Group Member and vice chair of the working group "Information systems for interconnected logistics" of the European Technology Platform ALICE. He has participated in several FP7 and H2020 European research projects.

List of up to 5 relevant publications, and/or products, services, or other achievements relevant to the call content
• Wenzel, Hannah; Smit, Daniel; Sardesai, Saskia. A literature review on machine learning in supply chain management. In: Proceedings of the Hamburg International Conference of Logistics (HICL). epubli GmbH, 2019. S. 413-441. • Moroff, Nikolas Ulrich; Sardesai, Saskia. Machine learning in demand planning: cross-industry overview. In: Hamburg International Conference of Logistics (HICL) 2019. epubli GmbH, 2019. S. 355-383.
List of up to 5 relevant previous projects or activities, connected to the subject of this proposal
<u>European Projects</u> • European project CSA NextNet – Next Generation Technologies for Networked Europe (https://nextnetproject.eu/) provided a strategic research agenda for the development of future supply chains • Transforming Transport is a Big Data Value Lighthouse project for more efficient and sustainable transport (https://transformingtransport.eu/) <u>National Projects</u> • Innovationlab “Hybrid services for logistics” (founded by BMBF) • Competence Center Machine Learning Rhine-Ruhr “ML2R” for the development of modular ML applications (https://www.ml2r.de/en/landingpage/) • Machine Learning Research Clan (https://leistungszentrum-logistik-it.de/) Several industrial projects have been conducted in terms of local stock management and transfer to global inventory management. This includes a simulation study to streamline shipping structure and packaging lines.
Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work
Not expected

Partner	**IRIS Technology Solutions S.L.**		
Short Name	IRIS	**Type**	SME
Role	Partner	**Country**	Spain
Website	www.iristechnologygroup.com/	**Logo**	
Description of the legal entity			

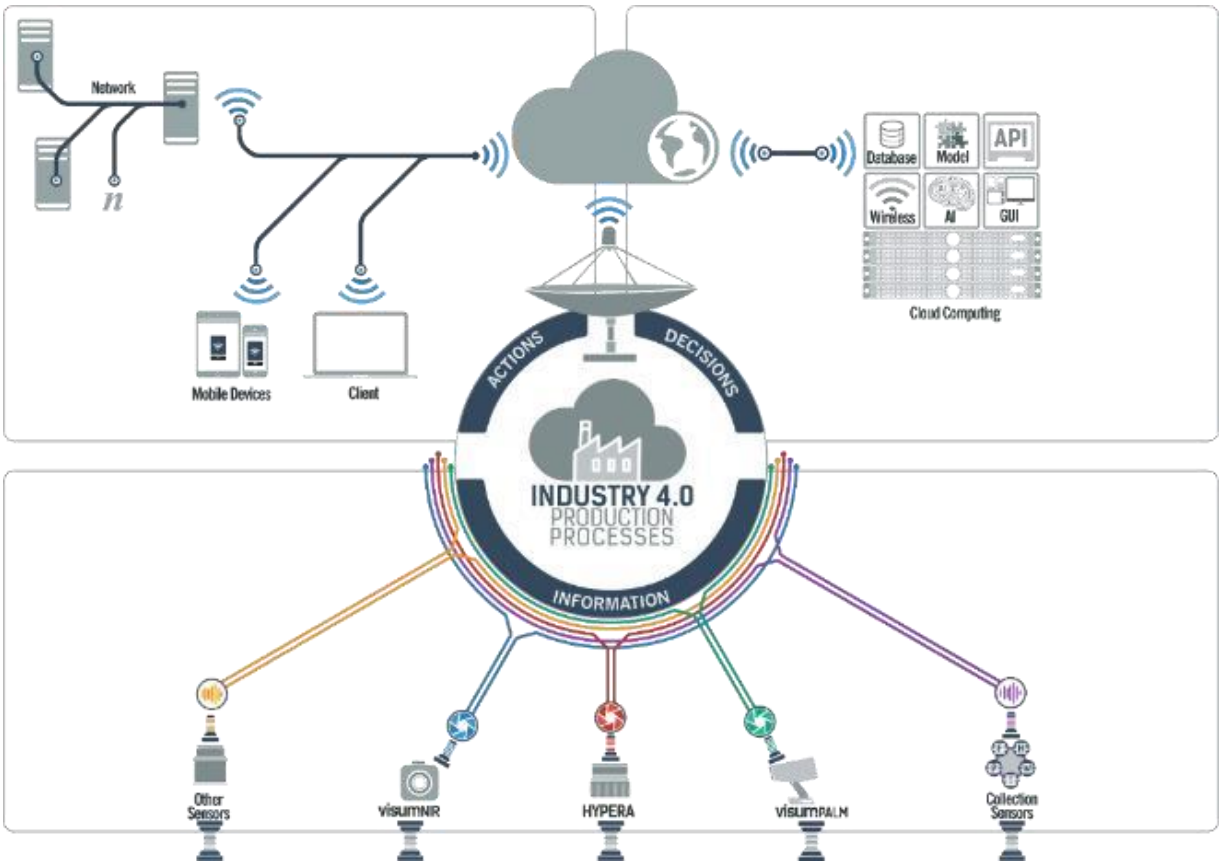
IRIS Technology Group (http://www.iristechnologygroup.com/) is an advanced engineering Group that was established in Barcelona in 2007. IRIS Technology Solutions S.L, located in Barcelona, specialises in the manufacture and integration of Process Analytical Technology (PAT)-based realtime quality monitoring solutions for the process industries and IoT systems for smart manufacturing, predominantly targeted at the food, pharmaceutical and chemical industries. In particular, our product line comprises proprietary state-of-the-art IR analysers for at-line and in-line process monitoring.

On the one hand, our solutions are based on the latest developments in photonics in the fields of MEMs, Near Infrared Spectroscopy, Hyperspectral Imaging, and Raman. On the other hand, we leverage ICTs, whereby we apply Machine learning, Artificial Intelligence and Data Mining techniques for predictive and active control, as well as for decision support systems. In addition to photonics-based analysers, we also work with IoT sensors for enabling equipment and machinery for smart manufacturing and transitioning industry to the Industry4.0 paradigm. Our solutions are cloudbased for mobility, remote monitoring and multi-site applications.

Our Innovation Division which is governed by an Innovation Committee comprising Senior Management, and an Innovation Unit for managing and executing the companies innovation roadmap, has 3 core strategic lines, for shaping the Factory of the Future:

1. New optical photonics-based technologies and PAT systems for production process optimisation
2. ICTs (Artificial intelligence / machine learning, IOT & cloud platforms)
3. Circular and Bio-based Economy: digitalization and waste valorisation

In addition to running an internal R&D programme, as well as carrying our various joint private projects with industry partners, our Innovation Division is highly experienced in working in collaborative R&D projects in H2020 (both as coordinators and partners), principally in the SPIRE, FoF, and BBI programmes. The company was established in 2007 and currently employs 60+ staff with a turnover of € 5 M. It is equipped with engineering workshops (3), food testing and wet chemistry laboratory, optics labs (2), and an electronics/telecommunications lab. IRIS is ISO 9001:2008 certified and also is certified under UNE-EN-ISO 166002: Management Systems R&D&i. We are also certified by the local (Catalonian) government as a provider of technological R&D services (Tecnio Certification). IRIS is a member of the following sectorial networks: SPIRE, SPIE and SECPHO, in addition to regional industrial collectives such as Smart Space (association of companies and organisations that specialise in the ‘smartification’ of industry).



Main tasks and responsibilities within the project

IRIS will lead WP1 where they will evaluate the current landscape for AI and BD technologies presence in industry, mapping the existing technologies, their applicability in process industry environments and their current uptake in different process industries. Coming from this status analysis, a plan will be developed to generate relevant impact on the 5 selected macro application areas.

IRIS is an active member of A.SPIRE with a relevant involvement in coordinating and participating in process and manufacturing innovation projects in H2020 framework. IRIS has a matrix of multidisciplinary knowledge groups in the company with strong expertise in engineering and science. With its team and resources, including electronics, mechatronics, mechanical, optical, industrial design engineers, as well as specialists in systems engineering, application programming and web development. The group can carry out full system design and engineering, including electronic design and assembly, firmware and software development, electrical harnesses and interconnections, mechanical design for enclosure and prototype integration, as well as installation, validation and industrialisation of prototypes and equipment in production plants. In addition to this broad crossdisciplinary engineering team, IRIS' research team consists of data scientists, chemometricians, chemists and physicists, among others, who have the expertise to validate the technologies developed.

Profiles of persons primarily responsible for carrying out the proposed activities

Elodie Bugnicourt	**Profile**	https://www.linkedin.com/in/elodie-bugnicourt-696a881/

Dr. Ing. ELODIE BUGNICOURT (female) Project manager and Leader of the innovation unit in IRIS since 2009. Masters in Material Science & Engineering (2002), PhD in Polymers and Composite Materials (2005), both from National Institute of Applied Science (INSA), Lyon, France. Coordinated several European projects in the fields of bio-based packaging materials development, process monitoring, surface treatment. Previous industry experience: Arkema; Wacker Chemie, Ashland.

Albert Torres	**Profile**	https://www.linkedin.com/in/albertorres/

MSc ALBERT TORRES. PMO Manager (Male) at IRIS. Masters in Industrial Engineering (2009) from the Polytechnic University of Catalonia (UPC-ETSEIB) in Barcelona. Postgraduate in Project Management (2013) from the University of Barcelona (UB-IL3) and currently developing a Postgraduate in Business Innovation, from the University of Catalonia (UPC). He spent almost 5 years as being member of the International Project Office of LEITAT Research Centre, where he could deployed the knowledge and experience of managing and coordinate Public Funded Research Projects under the different financial International, National and Regional schemes: H2020, ERA-Nets, JTIs, CIEN, RIS3CAT,… sharing experiences with the top R&D performing entities around Europe. At IRIS, he is leading the PMO Office with a team of five Project Managers, leading them with passion and optimistic attitude, believing that great results are achievable in well-functioning teams. Moreover, contributes on the business strategy development and commercial decision-making, sharing his experience in risk management and portfolio management.

David Nettleton	**Profile**	https://www.linkedin.com/in/dnettleton/

Dr. DAVID NETTLETON. (male) Data Mining Analyst. Has a Ph.D in Artificial Intelligence from the Universitat Politécnica de Catalunya, Barcelona, Spain, M.Sc. In Computer Software and Systems Design from the University of Newcastle upon Tyne, U.K., and B.Sc. (Hons) Computer Science from the University of Wolverhampton, U.K. Has over 20 years experience of applying data mining algorithms (including neural networks, rule induction) and statistical techniques (including ANOVA, regression, principal components) in a diversity of data analysis and modeling projects, such as medical diagnosis/prognosis, fraud detection, customer loyalty and segmentation, online user web search behaviour and data privacy. As well as commercial projects, he has conducted research and lectured at two universities. He has published a diversity of articles in specialised data analysis journals/congresses and three books. He has participated in several competitive multi-partner research projects and has a European patent application for a graph matching algorithm.

Ignacio Montero	**Profile**	http://www.innovationatiris.com/capacities/

MEng IGNACIO MONTERO CASTRO (Male). Project and Innovation Manager. Chemical Engineer from University of Santiago de Compostela, Spain, and Master in Process Engineering from Polytechnic University of Catalonia, Spain. Ignacio comes from a background in process control and industrial safety, developing applied research in selfoptimizing control strategies for industrial facilities and environmental risk analysis for the Catalan department of Industry. He spent almost 3 years on the Circular Economy unit of LEITAT Technological Centre working as R&D engineer and technical project manager. During this period, he was responsible for management, execution and proposal writing of Public Funded Research Projects under the different financial International, National and Regional schemes: H2020, CIEN, LIFE. Since joining IRIS in July 2018, he has been providing business management services to support the award, coordination and implementation of EU funded projects on Applied Photonics, ICT & Digitalisation and Circular & Bio-economy sectors.

List of up to 5 relevant publications, and/or products, services, or other achievements relevant to the call content

- Predicting rice blast disease: machine learning versus process-based models. Nettleton, D.F., Katsantonis, D., Kalaitzidis, A. Sarafijanovic-Djukic, N., Puigdollers, P., Confalonieri, R. BMC Bioinformatics 20, 514 (2019) doi:10.1186/s12859-019-3065-1
- A. Rosales, O. Mc Nerney. Sensing Technology. Chapter 6. Resource Efficiency of Processing Plants. Monitoring and Improvement. Edited by Stefan Krämer and Sebastian Engell. Wiley-Vch. Print ISBN: 978-3-527-34074-3
- Relevant product lines: Monitoring devices for real-time quality control: ○ Visum PALM: NIR technology for incoming material quality control and detection of adulteration.
  - Visum In-line In-Line NIR monitoring for product standardization and increase of production efficiency.
  - HSI: Hyperspectral imaging solutions for the industry
- B2B contracts in the field of process industry monitoring and digitalisation.

1. Related services: Chemometrics model development and training.

List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

- **SPRING**, H2020 SPIRE, GA 767412, Setting the framework for the enhanced impact of SPIRE projects
- **SHAREBOX**, H2020 SPIRE, GA 680843, Secure Management Platform for Shared Process Resources
- **PROPAT**, H2020 SPIRE, GA 637232, Robust and affordable process control technologies for improving standards and optimising industrial operations
- **DIY4U**, H2020 FOF, GA 870148, Open Innovation Digital Platform and Fablabs for Collaborative Design and Production of personalised/customised FMCG
- **DATAMINE 4.0**, H2020 MSCA, GA 752907, Advanced Data Modeling and Analysis Applied to next Generation Industry 4.0 settings and the Internet of Things

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Not expected

Partner	National Council of Research		
Short Name	CNR-STIIMA	Type	Research Organization
Role	Partner	Country	Italy
Website	www.stiima.cnr.it	Logo	STIIMA Sistemi e Tecnologie Industriali Intelligenti per il Manifatturiero Avanzato Consiglio Nazionale delle Ricerche

Description of the legal entity

CNR-STIIMA is the Institute of Intelligent Industrial Technologies and Systems for Advanced Manufacturing of the National Council of Research. Its strategic mission is the growth of competitiveness and sustainability of manufacturing in Italy and Europe through research and industrial innovation. The RTD activities of the institute aim at innovating products, processes and organisations in several research areas, by focusing its competences on following applicative areas: mechatronic knowledge-based systems, robotics, multi-layer adaptive control systems, microsystems, virtual prototyping and integrated simulation of mechatronic manufacturing systems and supply chain models.
CNR-STIIMA acts as a strategic player in the continuous process of definition of the European, National and Regional visions on the manufacturing sector, through the participation in initiatives and platforms devoted to the formulation of new paradigms in the area of Manufacturing. In fact, CNR-STIIMA is among the contributors who inspired the creation of the ManuFuture platform and is a founding member of the EFFRA association and contributes to the EFFRA Strategic Roadmap outlining the guidelines for European research and innovation activities in the discrete manufacturing sector.
CNR-STIIMA is also founder and is part of the management body of the National Cluster of Intelligent Factories in Italy (www.fabbricaintelligente.it), cluster of almost 300 organizations including research centers, industrial companies and associations.
At the moment, CNR-STIIMA has active more than 30 EU projects in the areas of FOF, NMBP, ICT, SPIRE and CIRC and more than 10 national projects.

Main tasks and responsibilities within the project

CNR-STIIMA is involved in AI-CUBE for its experience in roadmapping and forecasting developed in previous EU and national projects. During the project CNR-STIIMA will be involved in the activities related to the analysis and mapping of technologies and in particular into the definition of a assessment framework to evaluate the level of application of these technologies for the definition of a roadmap. The knowledge and background on Supply chain topics will be also used to developing project content in the macro-areas asked by the call.

Profiles of persons primarily responsible for carrying out the proposed activities

Rosanna Fornasiero	Profile	it.linkedin.com › rosanna-fornasiero-47a24614

Rosanna Fornasiero (Female) is a researcher at CNR working in the area of Supply Chain Management, Mass Customization, Innovation and industrial policies. She has experience in many EU projects (FP6, FP7 and H2020) and she has acted as scientific manager and project coordinator in EU FP7 and H2020 projects in the area of NMP, FOF, ICT. She is responsible of the Roadmapping group of the Italian Cluster of Intelligent factories (www.fabbricaintelligente.it). She works with industrial associations to support regional research projects and technology transfer process to SMEs. She has more than 60 publications in scientific journals and conference proceedings and she is part of the scientific committee of conferences in the IFIP are like PRO-VE (IFIP-WG5.5), APMS (IFIP-WG5.7), ICE conference.

Other CNR-ITIA staff members who will be involved in the project activities are industrial and management engineers experienced in research and industrial projects related to managerial implication of digitalization, models for collaborative networks, digital technologies.

List of up to 5 relevant publications, and/or products, services, or other achievements relevant to the call content

- Pessot, E., Zangiacomi, A., Berkers, F., Hidalgo-Carvajal, D., Weerdmeester, R., & Fornasiero, R. (2019, September). Investigating Supply Chains Models and Enabling Technologies Towards Collaborative Networks. In Working Conference on Virtual Enterprises (pp. 335-343). Springer, Cham. PRO-VE conference 2019, https://doi.org/10.1007/978-3-030-28464-0_29
- Zangiacomi A., Pessot E., Fornasiero R., Bertetti M.& Sacco M. (2019) Moving towards digitalization: a multiple case study in manufacturing, Production Planning & Control, DOI: 10.1080/09537287.2019.1631468
- Zangiacomi A., Oesterle J., Fornasiero R., Sacco R. & Azevedo A. (2017) The implementation of digital technologies for operations management: a case study for manufacturing apps. Production Planning & Control Vol. 28 , Iss. 16, 2017.
- Camarinha-Matos, L. M., Fornasiero, R., & Afsarmanesh, H. (2017). Collaborative Networks as a Core Enabler of Industry 4.0. In "Collaboration in Data rich environment" (pp. 3-17). Springer.
- Macchion, L., Fornasiero, R. and Vinelli, A., 2016. Supply chain configurations: a model to evaluate performance in customised productions. International Journal of Production Research, pp.1-14.

List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

STIIMA-CNR Scientific and Technological background and excellence on topics related to the main tasks in AI-Cube project is grounded on the involvement in several European projects in the area of supply chain, digital technologies as well as roadmapping and forecasting. The following list represents the most recent projects related to AI-Cube topics both from the technological and the organizational point of view and are:

- 2017-2019: **NEXT-NET** (NMPB37-2016-CSA: Next generation technologies for networked Europe). Role: Project coordinator and principal investigator. Most important scientific activities: development of a strategic research agenda for supply chain of 2030; future scenarios definition, future strategies to integrate smart manufacturing and logistics processes.
- 2016-2017: **INSPIRE** (SPIRE-06-2016-CSA: Business models for flexible and delocalized approaches for intensified processing). Role: partner. Most important scientific activities: roadmap for the Integration of process industry and discrete manufacturing through innovative technologies. Definition of supply chain models for process industry.
- 2017-2019: **DISRUPT** project (http://www.disrupt-project.eu/): Decentralised architectures for optimised operations via virtualised processes and manufacturing ecosystem collaboration, funded under FOF11-2016 H2020. Main activities are related to development of CPS system with industrial application in automotive and white goods sectors. Moreover, development of inbound logistics management tool for automotive sector.
- 2014-2017: **APPS4aME** project (FoF.NMP.2012-6: Engineering Apps for advanced Manufacturing Engineering) Main actitivies: tools and methods for mobile applications in the manufacturing sector.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Not expected